

Unit 8 Multiple Choice Practice

1. How many liters of distilled water must be added to 10. mL of an aqueous solution of HNO_3 with a pH of 2 to create a solution with a pH of 3?

a. 10. mL

b. 20. mL

c. 40. mL

d. 90. mL

2. The $[\text{OH}^-]$ in a solution with a pH of 3.00 is

a. $1.0 \times 10^{-11} \text{ M}$

b. $1.0 \times 10^{-9} \text{ M}$

c. $1.0 \times 10^{-6} \text{ M}$

d. $1.0 \times 10^{-3} \text{ M}$

3. Consider the following:

I. H_2SO_4

II. HSO_4^-

III. SO_4^{2-}

Which of the chemical species above are present in a reagent bottle labeled 1.0 M $\text{H}_2\text{SO}_4(\text{aq})$?

a. I only

b. I and II only

c. II and III only

d. I, II, and III

4. The pH of a 1.0 mL solution of 0.30 M NaOH if 9.0 mL of distilled water is added to the solution is closest to:

a. 3

b. 11

c. 12

d. 13

$$M_1 V_1 = M_2 V_2$$

$$0.30 \text{ 1 mL} = M_2 \cdot (9+1)$$

$$M_2 = 0.03 \text{ M}$$

$$-\log 0.03 = 1.52$$

$$\text{pH} = 14 - 1.52 = 12.48$$

5. A 0.25 M solution of a weak monoprotic acid has a pH of 6. What is the value of K_a for the acid?

a. 4×10^{-12}

b. 4×10^{-11}

c. 2.5×10^{-6}

d. 2.5×10^{-5}

$$K_a = \frac{a^2}{HA} \approx \frac{(1 \times 10^{-6})^2}{0.25} = 4 \times 10^{-12}$$

6.. Consider the following equilibrium: $2 \text{H}_2\text{O}(\text{l}) + \text{energy} \leftrightarrow \text{H}_3\text{O}^+(\text{aq}) + \text{OH}^-(\text{aq})$

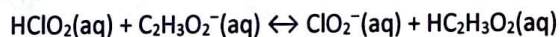
Which of the following describes the result of decreasing the temperature?

- | | $[\text{H}_3\text{O}^+]$ | $[\text{OH}^-]$ | K_w |
|----|--------------------------|-----------------|-----------|
| a. | increases | increases | increases |
| b. | decreases | increases | decreases |
| c. | increases | decreases | no change |
| d. | decreases | decreases | decreases |

run shifts left!

7. The pH of a solution changes from 3.00 to 6.00. By what factor does the $[\text{H}_3\text{O}^+]$ change?

- a. 3 b. 30 c. 100 d. 1000



8. The equilibrium constant for the reaction represented by the equation above is greater than 1.0. Which of the following gives the correct relative strength of the acids and bases in the reaction?

- | Acids | Bases |
|--|---|
| a. $\text{HC}_2\text{H}_3\text{O}_2 > \text{HClO}_2$ | $\text{ClO}_2^- > \text{C}_2\text{H}_3\text{O}_2^-$ |
| b. $\text{HClO}_2 > \text{HC}_2\text{H}_3\text{O}_2$ | $\text{C}_2\text{H}_3\text{O}_2^- > \text{ClO}_2^-$ |
| c. $\text{HClO}_2 > \text{HC}_2\text{H}_3\text{O}_2$ | $\text{ClO}_2^- > \text{C}_2\text{H}_3\text{O}_2^-$ |
| d. $\text{HC}_2\text{H}_3\text{O}_2 > \text{HClO}_2$ | $\text{C}_2\text{H}_3\text{O}_2^- > \text{ClO}_2^-$ |

$K > 1$
more products!

If HClO_2 is stronger
then CB (ClO_2^-)
is weaker!

9. When 10.0 mL of 0.10 M HCl is added to 10.0 mL of water, the concentration of H_3O^+ in the final solution is:

- a. 0.010 M b. 0.050 M c. 0.10 M d. 0.20 M

10. A 0.50 M solution of a weak monoprotic acid has a pH of 4. Calculate the ionization constant, K_a , for the acid.

- a. 5×10^{-9} b. 2×10^{-8} c. 2×10^{-4} d. 5×10^{-5}

$K_a \approx \frac{a^2}{HA} = \frac{(1 \times 10^{-4})^2}{0.50} =$

11. A bottle of water is placed in an ice bath to chill. What happens to the pH of the water as it cools?

- yes! more H⁺, less H⁺, increase pH*
- a. It will increase because the auto-ionization of water is an endothermic process. *would decrease if endo*
 b. Nothing; the volume would have to change in order for any ion concentration to change.
 c. Nothing; pure water always has a pH of 7.00.
 d. It will decrease because the concentration of H⁺ is decreasing.

) if T changes, K changes!

12. In order to change the pH of a solution from 2.0 to 4.0 the [H₃O⁺] must

- a. increase by a factor of 2
 b. decrease by a factor of 2
 c. increase by a factor of 100
 d. decrease by a factor of 100

13. What is the conjugate acid and the conjugate base of HPO₄²⁻?

Conjugate Acid

Conjugate Base

- | | |
|--|---|
| a. PO ₄ ³⁻ | H ₂ PO ₄ ⁻ |
| b. H ₂ PO ₄ ⁻ | PO ₄ ³⁻ |
| c. H ₂ PO ₄ ⁻ | H ₃ PO ₄ |
| d. H ₃ PO ₄ | PO ₄ ³⁻ |

14. When comparing 1.0 M solutions of bases, the base with the lowest [OH⁻] is the

- a. weakest base and it has the largest K_b value.
 b. strongest base and it has the largest K_b value.
 c. weakest base and it has the smallest K_b value.
 d. strongest base and it has the smallest K_b value.

15. What is the concentration of Sr(OH)₂ in a solution with a pH = 11.00?

- a. 2.0 × 10⁻¹¹ M b. 1.0 × 10⁻¹¹ M c. 5.0 × 10⁻⁴ M d. 1.0 × 10⁻³ M

[H⁺] = 1 × 10⁻¹¹, [OH⁻] = 1 × 10⁻³

1 × 10⁻³ OH⁻ / (Sr(OH)₂)₂

2 OH⁻

=

For which of the reactions above is the net ionic equation H⁺ + OH⁻ → H₂O?

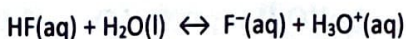
16. Consider the neutralization reactions between the following acid-base pairs in dilute aqueous solutions:

- (1) CH₃COOH + NaOH →
 (2) HNO₃ + KOH →
 (3) H₃PO₄ + Ba(OH)₂ →
 (4) HCl + KOH →
 (5) H₂SO₄ + NH₃ →

For which of the reactions above is the net ionic equation H⁺ + OH⁻ → H₂O?

- a. 1, 3 b. 2, 4, 5 c. 2, 4 d. 3, 5

strong acid + strong base only!



2. The ionization of HF(aq) in water is represented by the equation above. In a 0.0350 M HF(aq) solution, the percent ionization of HF is 13.0 percent.
- a. Two particulate representations of the ionization of HF molecules in the 0.0350 M HF(aq) solution are shown below in Figure 1 and Figure 2. Water molecules are not shown. Explain why the representation of ionization of HF molecules in water in Figure 1 is more accurate than the representation in Figure 2. (1 point)
- (The key below identifies the particles in the representations.)

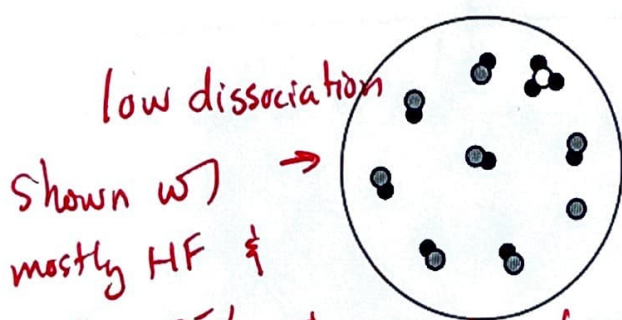


Figure 1

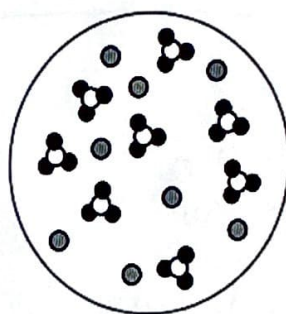


Figure 2

← indicates 100% dissociation b/c No HF remains, only F^- , indicative of strong acid.

- b. Use the percent ionization data above to calculate the value of K_a for HF. (2 points)

$$K_a = \frac{[\text{H}^+][\text{F}^-]}{[\text{HF}]}$$

$$= \frac{(0.00455)^2}{(0.0350 - 0.00455)}$$

$$= 6.8 \times 10^{-4}$$

$$\%D = 13\% = 100 \cdot \frac{[\text{H}^+]}{[\text{HF}]} = 100 \cdot \frac{[\text{H}^+]}{0.0350}$$

$$[\text{H}^+] = [\text{F}^-] = 0.00455$$

- c. If 50.0 mL of distilled water is added to 50.0 mL of 0.0350 M HF(aq), will the percent ionization of HF(aq) in the solution increase, decrease, or remain the same? Justify your answer with an explanation or a calculation. (1 point)

Increase. More dilute acid has higher % diss.

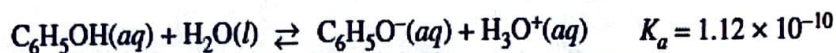
$$K_a = 6.8 \times 10^{-4} = \frac{[\text{H}^+][\text{F}^-]}{[\text{HF}]}$$

the 2x volume $\rightarrow (0.0350/2)$

$$[\text{H}^+] = 0.00345 \text{ M}$$

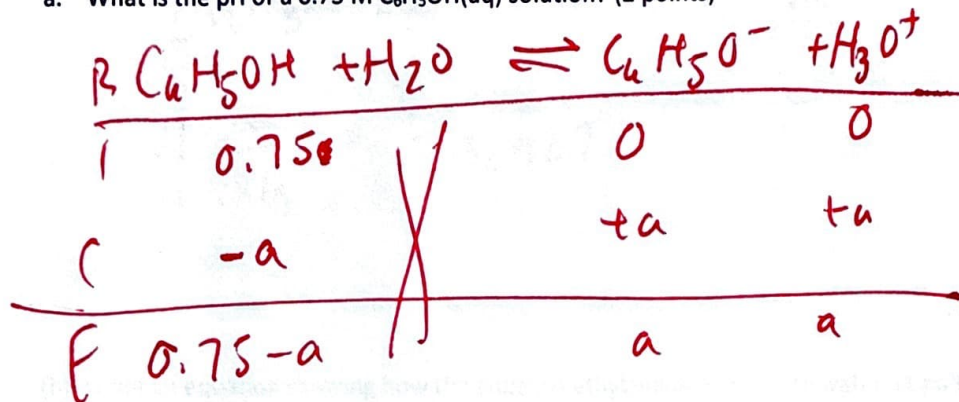
$$\%D = 0.00345 / 0.0175 = 20\% \text{ } \nearrow 13\%$$

FR Practice #3 (2016 #4, 4 points)



3. Phenol is a weak acid that partially dissociates in water according to the equation above.

a. What is the pH of a 0.75 M $\text{C}_6\text{H}_5\text{OH}(aq)$ solution? (2 points)



$$K_a = \frac{[\text{C}_6\text{H}_5\text{O}^-][\text{H}_3\text{O}^+]}{[\text{C}_6\text{H}_5\text{OH}]}$$

$$1.12 \times 10^{-10} = \frac{a^2}{(0.75 - a)}$$

a is
wee!

$$a = 8.854 \times 10^{-6}$$

$$\text{pH} = -\log [\text{H}^+] = -\log 8.854 \times 10^{-6}$$

$$= \underline{\underline{5.05}}$$

Free Response Practice #4 (5 points)

A pure 14.85 g sample of the weak base ethylamine, $\text{C}_2\text{H}_5\text{NH}_2$, is dissolved in enough distilled water to make 500. mL of solution.

- (a) Calculate the molar concentration of the $\text{C}_2\text{H}_5\text{NH}_2$ in the solution. (2 points)

$$[\text{C}_2\text{H}_5\text{NH}_2] = n/v = 0.32927 / 0.500 = \underline{0.659\text{M}}$$

$$\frac{14.85\text{g}}{45.1\text{g}} \times \frac{1\text{mol}}{1} = 0.32927$$

- (b) Write an equation showing how the aqueous ethylamine reacts with water. (1 point)



- (c) Write the equilibrium-constant expression for the reaction between $\text{C}_2\text{H}_5\text{NH}_2(\text{aq})$ and water. (1 point)

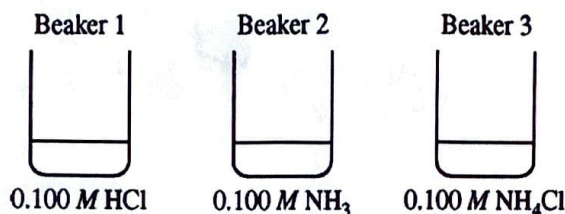
$$K_b = \frac{[\text{C}_2\text{H}_5\text{NH}_3^+][\text{OH}^-]}{[\text{C}_2\text{H}_5\text{NH}_2]}$$

- (d) Of $\text{C}_2\text{H}_5\text{NH}_2(\text{aq})$ and $\text{C}_2\text{H}_5\text{NH}_3^+(\text{aq})$, which is present in the solution at the higher concentration at equilibrium? Justify your answer. (1 point)

$$[\text{C}_2\text{H}_5\text{NH}_2] > [\text{C}_2\text{H}_5\text{NH}_3^+]$$

this is a weak base so $K_b < 1$
and only partially ~~dissolves~~ dissociates
ionizes. more weak base than conjugate
acid in solution

Free Response Practice #5 (2011 #1, 10 points)



2. Each of three beakers contains 25.0 mL of a 0.100 M solution of HCl, NH₃, or NH₄Cl, as shown above. Each solution is at 25°C.

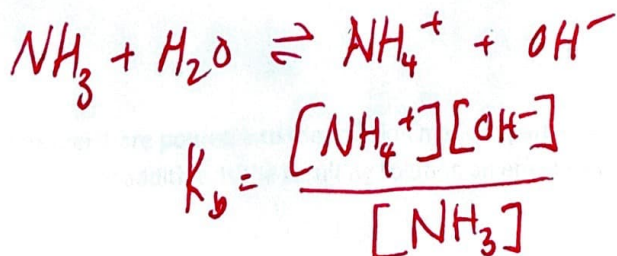
a. Determine the pH of the solution in beaker 1. Justify your answer. (1 point)

Strong acid!

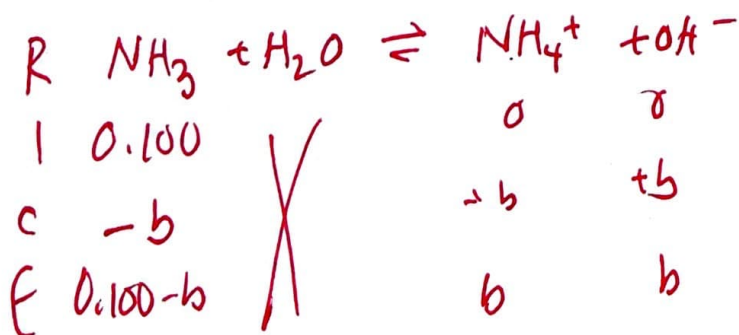
$$pH = -\log [HCl] = -\log (0.100) = 1.000$$

b. In beaker 2, the reaction $NH_3(aq) + H_2O(l) \leftrightarrow NH_4^+(aq) + OH^-(aq)$ occurs. The value of K_b for $NH_3(aq)$ is 1.8×10^{-5} at 25°C.

i. Write the K_b expression for the reaction of $NH_3(aq)$ with $H_2O(l)$. (1 point)



ii. Calculate the $[OH^-]$ in the solution in beaker 2. (2 points)



$$K_b = \frac{[NH_4^+][OH^-]}{[NH_3]}$$

bisweel.

$$\frac{b^2}{(0.100 - b)} = 1.8 \times 10^{-5}$$

$$b = 1.34 \times 10^{-3}$$

$$[OH^-] = 1.34 \times 10^{-3}$$

c. In beaker 3, the reaction $\text{NH}_4^+(\text{aq}) + \text{H}_2\text{O}(\text{l}) \leftrightarrow \text{NH}_3(\text{aq}) + \text{H}_3\text{O}^+(\text{aq})$ occurs.

i. Calculate the value of K_a for $\text{NH}_4^+(\text{aq})$ at 25°C . (1 point)

$$K_a(\text{NH}_4^+) = K_w / K_b(\text{NH}_3) = \frac{1 \times 10^{-14}}{1.8 \times 10^{-5}}$$

$$K_a = 5.56 \times 10^{-10}$$

ii. The contents of beaker 2 are poured into beaker 3 and the resulting solution is stirred. Assume that volumes are additive. Calculate the pH of the resulting solution. (2 points)

Buffer! Wait for 7.2

d. The contents of beaker 1 are poured into the solution made in part (c)(ii). The resulting solution is stirred. Assume that volumes are additive. Is the resulting solution an effective buffer? Justify your answer. (1 point)

Buffer! Wait for 7.2